

3 Yr. Degree/4 Yr. Honours 1st Semester Examination, 2024 (CCFUP)

Subject : Chemistry

Course: CHEM1011 (MAJOR)

(Basic Chemistry-I)

Time: 2 Hours

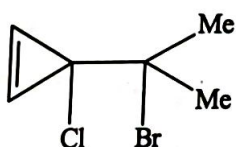
Full Marks: 40

The figures in the right hand margin indicate full marks.
Candidates are required to give their answers in their own words
as far as practicable.

1. Answer any five questions from the following:

2×5=10

- (i) Cite an example of a process where heat is completely converted into work.
- (ii) What is meant by 'Critical coefficient' of a gas?
- (iii) "Activation energy of some reactions may be zero."— Justify or criticise.
- (iv) Which one is more basic and why? MgO and Al_2O_3 .
- (v) What is 'Sacrificial hyperconjugation'? Give an example.
- (vi) Compare dipole moment between fluorobenzene and chlorobenzene.
- (vii) Identify the halogen atom to be precipitated in presence of $AgNO_3$ solution and why?



- (viii) An element with mass number 81 contains 31.7% more neutrons as compared to protons.
Assign the atomic symbol.

2. Answer any two questions from the following:

5×2=10

- (a) (i) The virial equation of state in terms of P is given by

$$Z = 1 + \frac{1}{RT} \left(b - \frac{a}{RT} \right) P + \frac{a}{(RT)^3} \left(2b - \frac{a}{RT} \right) P^2 + \dots$$

Set up an expression for the initial slope of Z vs P curve (at $P \rightarrow 0$) of a real gas and obtain the expression for Boyle temperature.

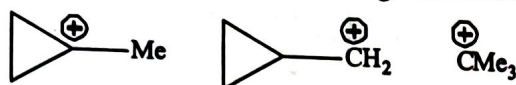
2+1

- (ii) Show that $PdV - VdP$ is not an exact differential.

2

- (b) (i) Draw π -molecular orbitals with electron occupancy of the allyl radical system. 3

- (ii) Compare the stability of the following carbocations: 2



- (c) (i) Arrange the following Lewis acids in order of increasing acidic character: 3
 BF_3 , BCl_3 , BBr_3

- (ii) The ionisation potential of H-atom is 13.6 eV. Calculate the ionisation potential of Li^{2+} ion. 2

- (d) (i) Is cyclo-octatetraene an aromatic compound?—Comment. 2

- (ii) In what condition $\text{pH} = \text{pKa}$ for an acidic buffer? 1

- (iii) Calculate the half-life of the decomposition reaction at 300K of a substance having initial concentration 0.05 mole L^{-1} . (Given, rate constant $= 6.8 \times 10^{-4} \text{ L mole}^{-1} \text{ s}^{-1}$) 2

3. Answer any two questions from the following:

10×2=20

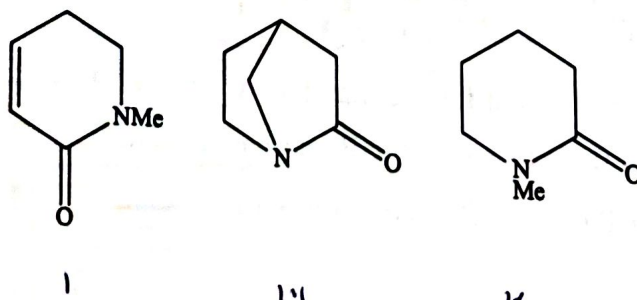
- (a) (i) The temperature variation of rate constant (k) of a reaction is given by $k = A e^{-E_a/RT}$ (Terms have their usual significance). Suggest a suitable plot for the determination of E_a . 2

- (ii) For a second order reaction, $A \rightarrow \text{Products}$, show that the time required for the three-fourths of the reactants to decay ($t_{3/4}$) is equal to $3t_{1/2}$, where $t_{1/2}$ indicates half-life. 3

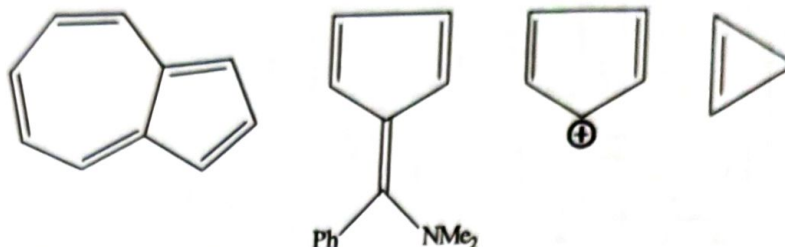
- (iii) Heat of neutralisation of HCl with NaOH is estimated to be -13.7 kcal . When 10 ml of (N/10) acetic acid is neutralised by 10 ml of (N/10) NaOH, enthalpy change is -12.5 kcal . Calculate the heat of dissociation of acetic acid. 3

- (iv) Show that $\left(\frac{\partial P}{\partial T}\right)_V = \frac{\alpha}{\beta} \left[\begin{array}{l} \alpha = \text{coefficient of volume expansion} \\ \beta = \text{coefficient of compressibility} \end{array} \right]$ 2

- (b) (i) Arrange the following compounds in order of increasing basic strength. Justify your answer. 3



- (ii) Categorize the following species as aromatic/anti-aromatic/non-aromatic with brief reason: 4



- (iii) C—C bond distance in CCl_3CHO is calculated to be $1.54\text{\AA}$ (same as ethane) while that in CH_3CHO is observed to be $1.50\text{\AA}$. —Explain. 3

3×2=6

- (c) (i) Write short notes on (any two):

- (I) Inert pair effect
- (II) Allred-Rochow's electronegativity scale
- (III) Drago-Wayland equation

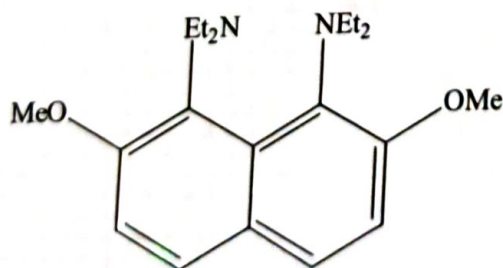
- (ii) Calculate pH of a 10^{-8}M aqueous HCl at 25°C . 2

- (iii) Should methyl orange be used as indicator in the titration of acetic acid by NaOH ? —Explain. 2

- (d) (i) What do you mean by 'The law of correspondig state'? Derive the necessary equation in this regard starting from the van der Waal's equation of state. 1+3

- (ii) Calculate the mass of a photon having wavelength $3.6\text{\AA}$. 2

- (iii) The following compound is a strong base. Explain why. 2



- (iv) Arrange the following ions in ascending order of radius: 2

